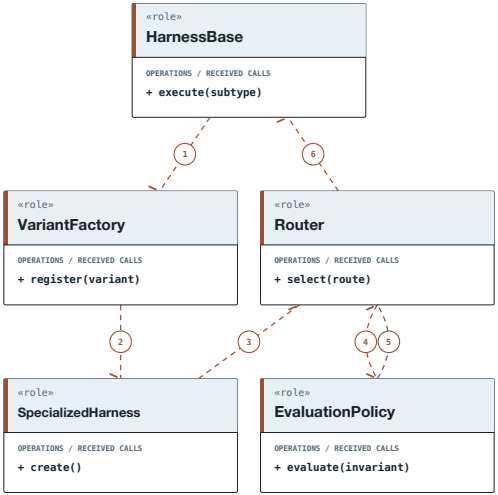
E01 Inheritance and polymorphic routing

Evolution / UML CLASS VIEW / ANALYTICAL MODEL

Specialize related harnesses.

UML CLASS VIEW

dashed arrow: call dependency · brown: return/failure · analytical, not an observed trace



POINTCUT / PATTERN INTERFACE

CONTEXT	Harness variants share a base abstraction.
TRIGGER	A base-typed slot dispatches to a concrete implementation.
STRUCTURE	Base interface, overridden behavior, subtype registry, and invariant clauses.
EFFECT	Any accepted subtype honors the base contract at the dispatch boundary.

CONTRACT / REQUIRED BEHAVIOR

REQUIRES	The base interface declares substitutable behavior.
ENSURES	Runtime dispatch reaches the intended specialized implementation.
INVARIANT	Subtype overrides do not narrow required capabilities or violate base obligations.
VIOLATION	Unsupported subtype behavior raises or is excluded from the slot.

RELATIONSHIPS / ANALYTICAL CALLS

number · caller → callee · message

1 HarnessBase → VariantFactory register(variant)
2 VariantFactory → SpecializedHarness create()
3 SpecializedHarness → Router select(route)
4 Router → EvaluationPolicy evaluate(invariant)
5 EvaluationPolicy ↩ Router allow_or_reject()
6 Router → HarnessBase execute(subtype)

VARIATION / TEST / NEGATIVE CONTROL

VARIATION	Class inheritance, interfaces, protocols, and plugin subclasses.
TEST_OBLIGATION	Test substitutability, dispatch, override errors, and invalid subtype.
NEGATIVE_CONTROL	A class hierarchy with no harness dispatch does not count.